ActInf Textbook Group ~ Cohort 5 ~ Session 1 (Onboarding) ~ 9/19/2023

TextbookGroup/ParrPezzuloFriston2022/Cohort_5 | Cohort 5 ~ Session 1 (Onboarding) ~ 9/19/2023 | 38:51

all right hello everyone this is the
first meeting of cohort five of the
textbook group on September 19th 2020 23
and thanks everybody for
saying hi
um I will
share my screen and we will be looking
at
the cohort five live meeting
page
so this is kind of like the starting
place for participating or joining
cohort five you can always click on
somebody's icon on the top right of Coda
and jump to the exact page they're at if
it's not clear from looking at the um
sidebar
so here we are in the first meeting and
after today the recording will be here
just like in cohort for
there's recorded videos
so I guess there's many places to
begin but is there any question that
anybody wants to ask about any aspect of
the textbook group
thank you
how do we even
start
and evolve it each time and and
make it
fit
for everybody who's
doing this on their own time and who's
coming at it from different angles
Varun and anyone else you can raise your

hand or however
well uh just a quick question um does
the textbook
um include going through some of those
uh extra notebooks some like active
inference coding like active influences
from scratch or some of the like
programmatic implementations or does
this mainly stay only on the theory
great question
um
the textbook itself
all of the code is in Matlab it's in
appendix C
and it is not a major part of the
textbook itself however in the textbook
group people are bringing resources and
other references
um in the page called code
you'll find several dozen
implementations of active inference code
and we can certainly
arrange sessions like just us informally
or trying to have the authors or other
experts join to look over
some code
um features if people are interested in
this
but it's not a necessary or central part
of the textbook
however because the second half of the
textbook is oriented towards application
especially
than talking about how you go from like
a sketch of a system of Interest
and then use the recipe in chapter six
towards actually designing the
generative models so for people who are

already and wanting to actually be
working in code then it's there
and we can work on it
um s mail
yeah I asked question I wanna know
um clearly about one
very important term
supervised and the relationship with
free energy
that can help me for good understanding
this textbook I think these two tell me
is so important
surprise
yeah
yes great question so
look
many questions will arise
about all kinds of terms and about
relationships so here's just one way
that that this can be approached so I
went to the questions page
in the search box I'll type in Surprise
now this is going to bring up a lot of
questions because it's a very common
term
but
um we could review
the questions that have been asked
and see if if one of them
directly asks what you're looking for
and if not then
it's like
a huge Act of service to add the
question
because it's something that we're all
curious about and so this is how we kind
of build our repertoire of
asking questions and also Everyone is

always welcome on their own time to go
into the answers and discourse section
and um develop
the answer whether by adding material or
by reorganizing
shortly to answer your question directly
surprise is about how expected or
unexpected a given sensory observation
is so if you were least surprised by a
sensory observation you would have had
like the most accurate belief and so
surprise minimization is equivalent to
having the model with a maximum evidence
surprise in and of itself though is
often very hard to calculate and so
instead of calculating surprise directly
attractable optimizable bound on
surprise is used and that's called free
energy
so that's definitely one of the key
pieces which is like if we had the
minimum surprise model
we'd have the maximum evidence model
in the exact Bayesian case that would be
totally fine however in the high
dimensional setting it's not tractable
to compute surprise
and so we have this
optimizable heuristic
with free energy and reducing the KL
Divergence
Susan
yeah I'm I'm particularly interested in
Apple actually mapping context so is
there is there a a place that maybe
scenarios are stored
that you know that have been mapped or
could be mapped

it's a um agree
question there
if it comes up in the context of a
chapter
you can always
um beneath the the questions in each
chapter page definitely people have
talked about
um scenarios
and then also I will just add it to the
um one notes Here
but in the octave block friends project
we have several dozen settings for
cognitive modeling that people have
worked to map so I'll just add this link
for settings for cognitive modeling
right thank you
cool anyone else have any just random
thoughts where we can explore a few
other pieces
yeah I
am wondering given that you've
now run several
uh cohorts for the textbook group
um and so the knowledge base has already
been established but and you've had a
few chances to
uh to try this out I'm wondering what
you're hoping each cohort will bring to
either to the knowledge base or or how
do you see each cohort evolving the the
study of of these topics
good question definitely remains to be
enacted
um I'll just give thoughts but not in
any you know ranked or fine-tuned
version first I hope that everyone who
comes into the space synchronously or

asynchronously can
find epistemic and pragmatic value and
learn about what's meaningful to them
and be really in the game in the arena
with these topics as they're being
developed
that would be a great first thing and
then for people who want to
pick up litter in the environment
and paint a nice wall and add a little
bit of information on their way
it's
unbelievably helpful for people to take
an active stance towards their
engagement with Dakota it can look like
there's a lot
it's not even a thimble in the ocean
and so people who really take an active
stance towards looking at the material
and where there's a question that
doesn't have an answer just throw
something out there
and when there's a question that has a
bunch of information
just delete an empty row there's some
low-hanging fruit
that simply
inevitably improves the experience for
everybody
there's always increasing amounts of
augmented intelligence tools and
language models including with Coda that
we can play with and so on so even just
adding
small amounts of information or
resources that people find relevant
adding new questions just
helping

enact
in a body that there isn't a final state
of the knowledge base again this is like
a larger shared
document than any of us would have made
by ourselves on this textbook
but there's so many more directions to
go
so
I'm always kind of looking and speaking
towards people who see affordances to
improve what we have here
and just do different things with it and
there's just there's there's so so many
ways to go there's so many unstructured
notes
for every
there's and again there's fun things to
do with code AI
there's fun things to do with synthesis
with linking to other work
um in future textbook groups page
there's there's um
hundreds of lines of notes on
different thoughts that people have had
about how to make the material more
accessible and applicable and rigorous
just
enjoying it wherever you're at as you're
going through the group
and then if people would like to rejoin
for
future cohorts for this textbook group
or for other projects at The Institute
and
do more that's great
excellent thank you as a short follow-up
is there anywhere

uh either on Dakota or
some email link where people can
register themselves publicly
uh like a little database of people who
are participating in the textbook group
maybe as a way to advertise that I'm
I'm participating in this and and if
anyone else wants to reach out to me
because I'm working on something that
maybe you find interesting here's my
contact
s sure
um so one option for that is the
projects page
here's is a kind of Project based
um another option I'll just make a um
I'll put it on this overview page
I'll just make a table for a cohort five
people
so if people want to add their own
information they they can
so people can at the bottom of the
cohort five this is always a fun plot to
look at as well
on the x-axis it's people's like
self-evaluated familiarity with active
inference from not familiar to very
familiar
and then the y-axis is the Computer
Science Biology and math
familiarity
it's all over
I guess one of the only trends that I
see is that very few people
say that they have higher than 80
familiarity with active
of course all these self-evaluation
numbers who knows what they really mean

this reflects our
textbook group
through time so
people have just different backgrounds
in math and computer science biology and
all these things philosophy that that
weave together
so just staying in the game and where
there's uncertainty
like externalizing it with asking it
will keep your cognitive load light and
contribute to open science so that's
huge and fun
great thank you
any other
thoughts or questions
uh I have one question regarding the
structure of these ethics groups so uh
uh you have multiple of these cohorts
are there any particular structure in
terms of like organizing for instance
these uh weekly meetings uh like what to
do uh during this time uh was it
effective or uh is that basically um up
to us to decide on like um how do we see
I'm highly open to
people who want to explore different
formats
um you can look at every single meeting
to see exactly how it did happen
but that's not how it has to happen it's
not like we're locked in it's just that
we wanted to hit the ground running as
soon as the textbook was available in
2022 and then just keep this opportunity
around
um usually the way that we well earlier
on we were just simply grasping

with the textbook and so
we would often
scan through the chapter
like during the meetings so just working
through the chapter and talking through
the chapter
where I would really love to
see it is like people look at what
chapter we're reading in the coming week
so it's alternating times so usually not
everyone can go to both on the given
um and also the at the like you can go
to cohort four as well if you just want
to go to you know there's people are
retaking it we're all just taking it
multiple times
um so when the discussion starts
and again this can evolve but here's
just one way that we can hear a lot of
voices and get content and have like
spontaneous things too we'll go to
chapter one for example
and also I'll specifically look at the
cohort five questions
the it's all one underlying big table
just different views this is just to
make sure that if there's a new question
that cohort 5 asks that we'll be able to
get to it but like let's just like cover
it for you know
so then
start with the questions respects
the effort that the person put in ahead
of time
to surface their curiosity whether
they're at the meeting or not
and try to go through the questions that
are asked about the

text
and then if people have Curiosities
arising like during the meeting oh I'm
wondering now I'm wondering about 4.3
it's like just type it out
like if it's typed out it's forever and
then we have a place for the discourse
if it's just shot from the hip
it dissipates now we still it doesn't
dissipate nowhere because we still can
record it and obviously we publish the
transcripts of these textbook groups and
so on but like as Curiosities arise
if you add them to the questions table
and then we can structure the
discussions with just seeing with what's
interesting
going to questions that have already
been asked in the in
um certain settings in other times going
to new questions
and then in the time between the
meetings just people hopefully
putting in the time to
increase their situational awareness of
the coda itself
read through the textbook
it's many coats of paint situation
there are parts that are going to be
very dense with the use of the active
inference ontology or very technical or
things like that such as time boxing and
staying in the game and then yeah what
is the most effective synchronous
meeting strategy that's a good question
and then also we can make kind of
supplemental meetings for example
commonly some people want to do a math

learning group
that is like some people want to focus
more on the backgrounds
and like the basic math
like reading equations Bayesian
statistics and then other people want to
do
um work on like more of like connecting
the dots with with the difference
expressions
so if people want to do more math if
they want to do more programming if they
want to be developing their project you
know don't don't let the Institute be
the rate limiting step
this is all up to people in their own
Learning Journey and connecting to
people here
and um
if you stay in the game over the coming
three months
you will get a lot more familiar with
active inference even just with one hour
per week
if you do more than one hour per week
active inference
it's not all in the textbook
in this one 2022 short textbook
but this is a great
Landmark for the field as the first
textbook obviously
and it allows us to reference out
to a lot of other
typing Edge work
so the discussions are great for that
because we can kind of like be like oh
yeah there's this like live stream
happening tomorrow or here's what this

person said last week and so it lets us
kind of like
look at how it was stated in 2022 while
also bringing it closer to what's
ongoing in the field
and if people you know if in future
cohorts if people want to be a
facilitator it doesn't require technical
expertise inactive inference to be a
facilitator
that's an option there's many many
options
okay thank you for your
um detail um explanation I think uh yeah
I do get that uh there's plenty of room
for um
trying to uh
uh make this um a really approval
Journey yeah thanks
cool yeah
I've got a question
where where do you see
Innovative applications
that
may or may not be
utilizing active inference that you know
might signal us to kind of look at what
is coming out whether that be AI or you
know augmented negotiation or or you
know um uh social network analysis
is there any particular Cutting Edge
areas that are you know starting to get
attention that would be interesting to a
couple of us around
it's a good
a lot of applications happen in the
industrial settings so they're not
always

shared
like with the full methods and details
that we would expect from like a
scientific paper or like an open science
educational project so I think that's
that's one fundamental unknown part
at least in the learning setting
there's there's a lot of exciting things
with connecting
innovative ways of modeling
different systems like we're seeing
active inference and then
the application work remains octave
inference is a very
young fields
and so
[Music]
things that might be expected in a more
mature field are not commonplace at
and rather than interpret them as that
that as a dismal State of Affairs
people who find that exciting and want
to contribute there
will find that that even some
pretty
um conventional work can actually have
very impactful outcomes
um I'll I'll
just link a few things so here in the
bottom we have the active inference
ontology
so this is one of the
key features of what we develop at The
Institute
which is like we're speaking in English
right now but to a large extent you
could think of this as being a dialect
of English which is the active inference

ontology it's also been translated into
many languages by by experts
um and it's a project that we work on
every week with basically how to utilize
the active inference ontology and also
I'll add to this page the a knowledge
engineering link
we've then used the ontology to analyze
the literature
and look at how terms change in their
use per year
but overall
let's see if if which plot is relevant
here but overall
we are talking about the number of
papers being in the low hundreds
over the last
10 to 15 years
most of them happening more recently
many of them happening
through Carl friston at all but that
also being increasingly decentralized
just keep the finiteness in mind
and look for the
affordances to contribute because there
are many many
so let's look at just the overall
textbook
oh yeah we're just in the kind of the
dot zero the the background and just
talking about the group itself chapter
one
and chapter 10 are
textual overviews they have no equations
they're like kind of like bookends on
the book
chapter two and again in our between now
and December we're going to be going

through just chapters one through five
but if you want to be also going to
cover four and going through six or ten
you can't
um and the author is very clearly say it
the first half of the book is like
epistemic it's about learning active
inference
take the time to really learn what
active inference is it will make
applying it
easier when you understand what it
really is
and then chapters six through ten are
exactly about the recipe of how you
apply it
now it's not at the Hold Your Hand
guidebook stage yet but we connect a lot
of the Dots here and this is the
direction that we're going in chapter
one gives an overview and it
contextualizes active inference in
relationship to a lot of other areas uh
that that people are interested in that
they study
chapter 2 and chapter 3 are the low road
and the high road to octave inference
we're going to talk about this like a
ton but here's figure 1.2
here's the low road and the high road
and they both meet an active inference
the low road asks how
and it answers it with Bayesian
statistics
and the high road asks why
and the answer is basically
self-observation self-organization self
persistence

and so active inference brings together
this persistence imperative of the free
energy principle
High Road y with a Bayesian statistical
implementation approach that's the low
road
so that's chapters two and three the low
road how and then the high road why
and then we get to the essence of active
inference which is the generative model
the generative model of active inference
is like
what is being built
this is what you have in hand when you
have an active inference model no
generative model
just speculative
with generative model you're talking
about something specific
and so chapter four goes into some of
the technical details about what these
generative models are
building on
some of the precursor steps that were
covered in the low road and the high
chapter four generative models chapter 5
is going to focus on one of the areas
that has the most empirical evidence and
the most well-studied area which is
mammalian neuroscience
so as much of this work arose from human
neuroimaging in the 1990s and early
2000s
there's a lot of work on human cognitive
and Behavioral Science
and so that's the first half of the book
context
how and why

what
and then where has it really been
applied the most so far but
as we look at
um every single day
people are expanding the scope of
active inference applications every
single day
but the area it's been most studied is
in mammalian neuroscience
that's the first half of the book
it hits hard
but that's what we're going to get to by
December and then just briefly chapter
six is a recipe for Designing active
inference models so it's kind of like a
procedure and a set of questions that
you can ask yourself as you're designing
a generative model
which is an important
rite of passage in people's work in this
area they'll talk about that in chapter
10.
chapter seven and eight focus on two big
families of generative models which is
generative models that treat time in a
discrete setting
like click click click click and then
models that treat time continuously like
a flow
chapter nine looks at the integration of
empirical data with generative models
model based data analysis and then
chapter 10
like chapter one provides context and
revisits a lot of the more conceptual
questions
given

what has been covered
so that's the main
outline of the book
and they kind of you know the places
that we'll go
but also
because our the questions take us
everywhere
and people always raise questions at
many many levels
so if there's like you know if there's
if there's an equation or if there's
some section that doesn't make sense or
it's not what you would have written or
you don't understand how they got from
here to there that's totally fine
I really encourage you just to continue
working through it and just pass through
the part slightly
you don't
feel expert on in the moment of your
first reading
or your Hunter 3D
any other um thoughts or questions
but yeah it should be really fun we are
going to
head into for the coming two weeks into
chapter one
so you can
and and you can ask in the chapter one
questions you can just
add them here
into the cohort Five Section then
they'll be tagged with with cohort five
um you can go to just the the questions
overall like chapter one questions
overall
and um it would be super super helpful

if you upvote the questions that you
find interesting
that helps prioritize them for the
discussion
and then also once we have questions
like maybe you know 10 or more or some
number of upvotes
and then we feel like there's some
questions where it's a very open
discourse but there's other questions
where it's kind of like there is an
answer
and in those situations
we can then go from
the question to
the curated
thoughts on the answer
and then we can translate that into
different languages we can make short
videos
we can make more bite-sized materials I
mean there's just many ways we can go
from there
like this is
this is the real work surfacing what
people with different backgrounds and
Curiosities are curious about
respecting and encouraging all such
voices
and then doing what we can
to coordinate and and
do what we can with the answers
not to shut the door because some of
these topics are are going to be just
huge
open inter-subjective questions
or it will just gesture to a whole area
of philosophy or some other area of

practice
but there's been a lot of great
questions so far
so there's a lot of value in Reading
what people have
written over the years
and reviewing the discourse and again
hopefully modifying it
if you're not sure
you can always like select text and then
click this little arrow and do like it
you know oh I don't think so if you want
to you can always do a suggest change
or you could add a comment you'd say oh
not sure about this shouldn't that word
be that
but if you're just adding a reference or
adding connecting the dots or
reorganizing the material that already
exists
just the more the better
the more eyes that are looking at it
and and
um taking a deep breath
and trying to evaluate it and and
interrogating it in the moment
that is so huge and if it does make
sense don't assume that it just made
sense to someone else and that you just
don't get it just
if it doesn't make sense then that's
your opening to
ask that question to write the follow-up
and this is a short video with Ali who's
not here now but he'll probably join us
some future times
if you listen to the recordings you'll
you'll quickly become a fan of all these

gentle scholarship
um but yeah we'll go into chapter one
and we'll have the two weeks to discuss
it which whichever one or both people
join at again chapter one's a very
um prose-based chapter it's not equation
heavy
it brings up huge ideas though
and so then we'll just cruise through
the discussion
on one here in the equation section
so we translated all of the equations
into latex plain text
and then
um here's using Coda AI
to explain to high schoolers write
pseudocode explain the variables
give philosophy
um questions translate into difference
languages
um code AI I believe currently uses gbt
3.5
but cert and then also perplexity.ai you
can upload a file so I often have just
uploaded the whole textbook
and had discourse
on the knowledge engineering project
again if people are interested in this
we are curating large amounts of the
active inference information
environments into fine-tuned language
models
with code and with visual recognition
eventually and some other features
so there's a lot of ways to go
so that's why we just stay in the game
have fun
okay any last comments

yes
exciting thank you Susan
all right
I will stop the recording but thank you
everybody till next time
thank you very much Daniel bye I really
appreciate
all this work that you're doing I'm
looking forward to the next meetings
thank you thank you see y'all